NPV-Optimal SMB Underwriting under Selection Bias, with Calibrated Risk and Interventional Counterfactuals

Team Vector Ventures

June 2026

1 Problem framing & assumptions violated

Every assumption a credit scorecard leans on—random labels, population overlap, a stationary window, honest inputs—breaks on this book; we name each break and respond to it rather than modelling straight through it.

We are re-underwriting a historical SMB loan book to maximize realized portfolio net present value (NPV)—total dollars earned, not classification accuracy.

Labels are missing-not-at-random behind a sharp gate. Outcomes exist only for loans the prior lender approved, and approval is, to numerical precision, a hard cutoff `prior_underwriter_score` ≥ 0.273: a propensity model separates approved from declined with AUC 1.00 and *zero* score overlap. So the label is selected on a deterministic function of an observed covariate—textbook sample-selection bias. Inverse-propensity reweighting is infeasible here by a positivity violation, *not by our choice*: a propensity model on the ordinary covariates alone still hits AUC 1.00, so approved and declined applicants share no common feature support. Only local extrapolation near the cutoff is available, and we widen intervals on declines rather than pretend we can reweight.

The split is forward in time. Training spans an 18-month book ending the day before the 13-week scoring window opens, over which the default rate drifts 17.5% $\rightarrow$ 20.6%. We learn default *timing* on the history but recalibrate the *level* in-window—the history tells us the shape, only the window tells us the rate.

Missingness is information, not absence. "Never declined elsewhere" (null) loans default at 14.3% versus 21.3% when present; no-bank-feed loans at 19.1% versus 16.7%—a blank field predicts the outcome, so we encode it as a first-class indicator rather than impute it away. **Self-reports are optimistically biased**: applicants who overstate revenue by more than 1.5$\times$ default at 38.6% versus 16.1%, so the *gap* between stated and bank-feed revenue—not the stated figure—is the signal, central to our causal treatment of `stated_annual_revenue` below.

2 Methodology

One discrete-time model serves all three deliverables: a calibrated probability that a loan defaults, times a per-band curve for when *it defaults, read out as an NPV sign for the approve/decline call.*

One shared discrete-time model. The brief's NPV depends on the *day* of default (a day-5 default loses roughly the principal; a day-55 default nearly breaks even), and Deliverable B is the cumulative-default curve. Both consume one timing model. We write each loan's cumulative-default curve as $F_i(t) = \mathrm{PD}_i \cdot S_{b(i)}(t)$, where $\mathrm{PD}_i = \Pr(y_i{=}1 \mid \mathbf{x}_i)$ is a calibrated probability of default by day 90 and $S_b(t)$ is a normalized timing shape ($S_b(90){=}1$) learned per credit band b. In words: *how likely* a default is, kept separate from *when* it arrives.

PD model. A five-fold `GroupKFold` (by `business_id`) LightGBM ensemble over 55 engineered features (out-of-fold AUC 0.774, Brier 0.117), isotonic-calibrated. Features are deliberately buffer-centric: because loans are $\sim$1.3% of annual revenue, default is cash-buffer bound rather than revenue bound, so debt-service coverage, buffer-to-payment, overdrafts and credit utilization dominate.

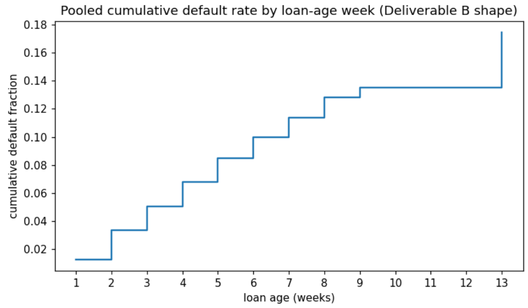

Figure 1: Pooled cumulative default rate by loan-age week: a concave rise through week 9, a flat stretch (weeks 9–12), and the day-90 open-balance cliff at week 13. Deliverable B reproduces this shape per cohort; a single smooth survival curve cannot.

The timing shape is bimodal and we model it as such (Figure 1). Of all defaults, 77.5% are missed-draw events over days 3–60; *no* defaults occur on days 61–89; then the remaining 22.5% spike exactly at day 90 (the open-balance sweep). A smooth survival fit would smear that mass across weeks, so we encode the empirical concave-rise/flat/day-90-cliff shape rather than blur the cliff. The shape is band-conditional—worse-credit borrowers default ∼13 days earlier and carry less day-90 mass—so $S_b(t)$ feeds both B and the per-loan $\mathbb{E}[t^*]$ used in A.

Deliverable A: decide on the NPV sign. We approve iff $\mathbb{E}[\mathrm{NPV}_i \mid \text{approve}] > 0$ using the exact product economics: revenue $F_i + R_i r(T/365)$ if repaid, and $F_i + D_i(t_i^* - 1) + \mathrm{rec}_i - R_i$ if default occurs at day t_i^*. That is, we keep the fee plus interest on a loan that repays, but only the fee plus the daily draws collected before t^* (less principal, plus $\approx 9\%$ recovery) on one that defaults—so a default costs less the later it lands. Because NPV is linear in t^*, the expectation uses the band-conditional expected default day. The break-even therefore sits at PD ≈ 0.39, not 0.5: even defaulters repay ∼74% of the schedule through daily ACH before defaulting, so a 39%-PD applicant is still NPV-positive and a 0.5 cutoff would leave money on the table. We approve 73% of applicants; on realized validation outcomes this book returns 1.74× the prior underwriter's P&L.

Deliverable B: shape × level. $\mathrm{CDR}_{w,a} = \mathrm{mean}_{i \in A_w} \mathrm{PD}_i \, S_{b(i)}(a)$—we average each approved cohort A_w's per-loan risk over the timing shape, giving a curve that only rises. Because the PD model has no cohort signal (training predates the window), we shrink each cohort's *level* toward the realized validation rate for the same calendar week (hierarchical-Bayes random effects; pseudo-count 75 = prior precision); validation and test span the identical 13 weeks, so this transfers.

3 Causal reasoning & counterfactual methodology

The judges' counterfactual queries ask what an intervention causes*, not what a feature* predicts*—so we classify every feature by whether editing it changes reality, and scale the effect accordingly.*

The counterfactual target is $p_q^{\mathrm{cf}} = \Pr\big(y_{i_q}{=}1 \mid \mathrm{do}(f_q{=}v_q), \mathbf{X}_{i_q,-f_q}{=}\mathbf{x}_{i_q,-f_q}\big)$ versus the observational $\Pr(y \mid f{=}v, \mathbf{X}_{-f})$: the first asks what happens if we *set* f to v, the second describes applicants who happen to *have* $f{=}v$, and they diverge exactly when f is confounded. Our **feature registry** keeps the edit honest: each engineered feature declares its parents, so setting a raw feature recomputes exactly its downstream features and nothing else—never an internally-contradictory applicant carrying a $20k loan with a $10k loan's payments. For example do(`requested_amount`) recomputes the daily-payment, buffer-to-payment, leverage and ask-to-revenue features; the provided `requested_amount_to_observed_revenue` is never reused (no stale-descendant bug).

We classify the intervenable features by causal status. *Mechanical causes* (`requested_amount`): changing the number changes the borrower's actual payment burden, so we perturb and propagate it through the registry at full strength. *Confounded proxies*—bureau and bank-feed symptoms of latent health (utilization, delinquency, revenue and cash signals)—are symptoms as much as causes: editing

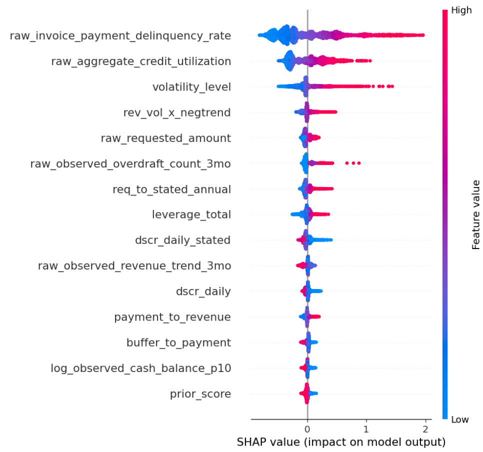

Figure 2: SHAP summary for the PD model: mean |SHAP| ranking with per-feature direction (red high / blue low). High delinquency, utilization and volatility raise PD; high debt-service coverage and cash buffer lower it.

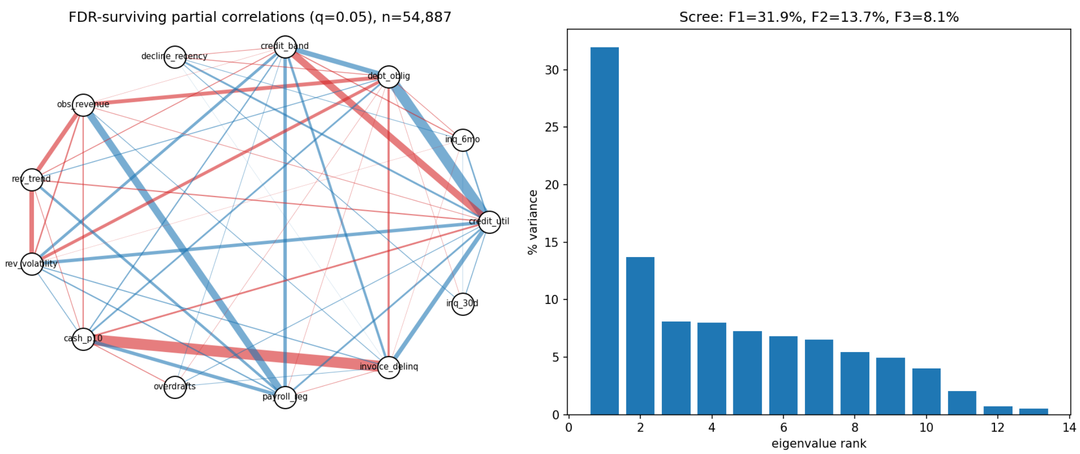

Figure 3: FDR-surviving partial correlations of the proxy block (q=0.05, n=55k; edge width = |partial corr.|, red = negative) and its eigenvalue spectrum—one dominant latent-health factor (32% of variance). The cash–invoice-delinquency edge (-0.78) is a descendant, not a sibling, so we exclude it from the adjustment set.

the thermometer does not cure the fever, so we keep only the estimated causal fraction $\hat{\lambda} = \beta_{\text{adj}}/\beta_{\text{naive}}$ (sibling-adjusted logistic; per-family $\hat{\lambda} \approx 0.52$, 0.51, 0.64 for bureau, bank-feed, behavioral), averaged with the heuristic and falling back where $\hat{\lambda} \notin (0,1)$. Conditioning sets are chosen *empirically* (Figure 3): an FDR-controlled sparse partial-correlation graph of the proxy block (q=0.05, n=55k) confirms one dominant latent-health factor (32% of variance, 2.3× the second) but flags "siblings" that are in fact *descendants*—invoice delinquency and overdrafts are consequences of low cash (partial corr. -0.78), payroll regularity of revenue (+0.50). Conditioning on a descendant blocks the true causal path, so we exclude them from those features' adjustment sets, correcting $\hat{\lambda}_{\text{cash}}$ from an implausible 0.03 (cash buffers mechanically fund the daily draw) to 0.44. *Self-reports*—`stated_annual_revenue`, `stated_time_in_business`—are returned at their observational PD: a different number on the form changes nothing about the business. Overstating revenue *predicts* default (38.6% vs 16.1%), but *forcing* a different statement causes nothing, so the interventional effect is ≈ 0—formally, $\Pr(y \mid \text{do}(\text{stated}), \mathbf{X}_{-\text{stated}}) = \Pr(y \mid \mathbf{X}_{-\text{stated}})$ when the claim itself does not cause default, the causally correct answer rather than naive re-prediction. *Immutable or proxy-by-fiat* features (`sector`, `geography_region`, `vintage_years`, `has_linked_bank_feed`) are not manipulable levers but are forced into the query set; we perturb the model, shrink the effect, widen the interval, and flag them as proxy effects.

Regulator defense. Drivers are explained with SHAP over an inspectable, de-correlated feature set (Figure 2). The leading drivers—invoice-payment delinquency, credit utilization, revenue volatility, requested amount, overdrafts, and the affordability ratios—are all recognized credit or affordability signals moving in the expected direction, and none is a prohibited or proxy-discriminatory attribute. Several queried values lie outside observed support (e.g. `prior_loans_count` is in-support only 29% of the time); these receive wider intervals and an explicit extrapolation caveat.

Model-class check. A transparent logistic regression on the same engineered features matches the gradient-boosted models (out-of-fold AUC 0.775 vs 0.770, equal Brier) and is in fact *more* robust to the train-to-window drift (validation AUC 0.759 vs 0.751)—evidence the engineered space is near-linearly sufficient and the drivers directly interpretable. Its standardized coefficients agree in sign with SHAP on eight of the top ten features; the two exceptions (`buffer_to_payment`, `debt_to_revenue`) are collinearity-driven sign flips—why we read drivers from SHAP, not raw coefficients. We rejected sequence and transformer architectures: no sequential structure, low-cardinality categoricals, discrimination data-limited at $\approx$0.77 AUC, and calibration—a scored component—favours the simpler, well-understood stack.

4 Calibration & uncertainty quantification

We fix the probability level *on the only data drawn from the scoring window, then size every interval by measured out-of-sample error rather than in-sample optimism.*

Point PD. Isotonic calibration is *refit on in-window validation* to correct the 17.5%→20.6% train-to-window drift (the $a_0 \to 0$ power-prior limit: in-window likelihood sets the level, history discounted); it stays valid on the approved manifold and is extrapolated on declines.

Intervals (A, C). Additive on the calibrated point, $p \pm \alpha z \sigma$: the point probability widened by how much the five folds disagree (σ), with the multiplier α fit by five-fold *cross-fit* on validation. Validated decile coverage is ≈ 0.90 at mean width ≈ 0.06, tighter than an earlier over-wide ($\approx$ 100% coverage) configuration—the brief's "contain the truth without being needlessly wide."

Trajectory intervals (B). A within-cohort bootstrap combined with a conformal band calibrated against the *true* validation cohort trajectories; validated coverage ≈ 0.92. Out-of-support counterfactual intervals in C are widened further (Table 1).

Deliverable	Metric	Value
A — PD model	out-of-fold AUC / Brier	0.774 / 0.117
A — policy	approve rate / P&L vs. prior underwriter	0.73 / 1.74×
A — intervals	decile coverage / mean width	0.90 / 0.06
B — trajectory	mean abs. error vs. realized CDR / coverage	0.013 / 0.92

Table 1: Validation evidence (in-window). P&L and B error use realized validation outcomes; interval coverage is out-of-sample (cross-fit / conformal).

5 Limitations & what we'd do differently

Our honest gaps trace to one fact—we never observe outcomes for declined applicants—so the decline and counterfactual estimates rest on extrapolation we cannot check.

Self-report and immutable counterfactuals remain heuristic (confounded proxies use the estimated $\hat{\lambda}$ above), and all C effects are unverifiable—the queried applicants have no labels.

Declines cannot be validated. Approved and declined applicants never overlap (§1), so reject inference is infeasible and only RD-style extrapolation remains (deferred); PD on declines—part of what A is scored on—stays unchecked.

Timing and cohort corrections lean on the validation window—they transfer only because validation and test share 13 weeks, though small per-cohort samples ($n \approx 150$) add variance. A is also marginally aggressive—a 20% PD shift still raises P&L—so a more conservative level would close the oracle gap.